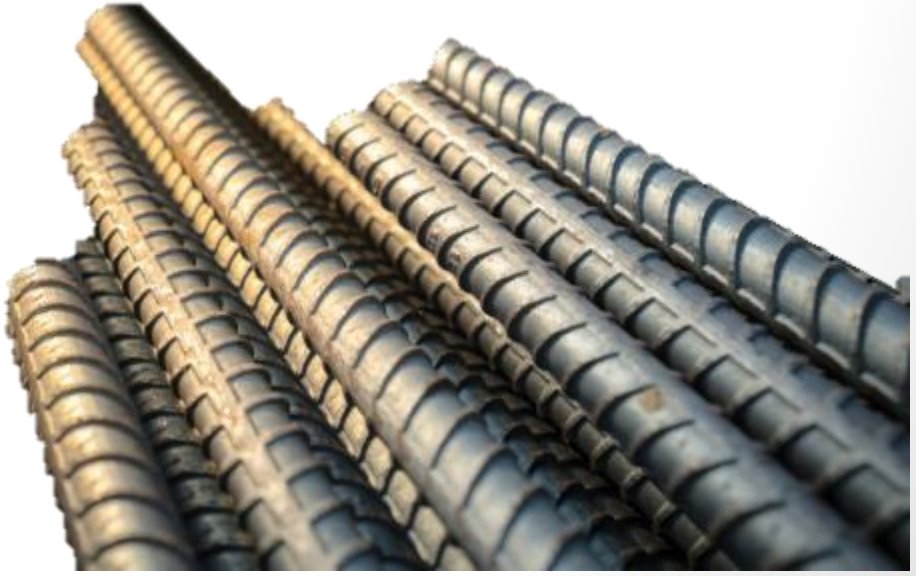
CE 208
Quantity Surveying

Reinforcement Estimation: RCC Slab



Necessity for Reinforcement Estimation

- For procurement of material
- To submit or to check at site the consumption of the material
- For preparation of bill for payment

Reinforcement used in RCC Work

- Plain round mild steel bar
- Deformed bar





Properties of ASTM Standard Reinforcing Bars

Bar No.	Diameter (in)	Nominal Area (in^2)
#3	3/8	0.11
#4	4/8	0.20
#5	5/8	0.31
#6	6/8	0.44
#7	7/8	0.60
#8	8/8	0.79
#9	9/8	1.00
#10	10/8	1.27
#11	11/8	1.56

Note: Nominal dimensions of deformed bar equivalent to plain bar; same weight per feet as plain bar

Reinforcement Estimation

1. By percentage basis of concrete
2. By thumb rule &
3. Accurate estimation based on detailed drawings

(1 & 2 are crude methods)

Reinforcement estimated in units of weight in Kilogram, Pound and Quintal (1 Quintal = 100 Kilogram)

Percentage Basis of Concrete

- Footing – 0.5% to 1%
- Column – 0.8% to 8%
- Beam – 1%
- Slab - 1%
- Lintel – 0.8%
- Sunshade – 0.5% to 0.8%



- Determine the amount of reinforcement in a RCC slab (20 ft x 15 ft x 5 inch) using percentage basis of concrete.

  (Unit weight of steel = 490 pcf)

Thumb Rule

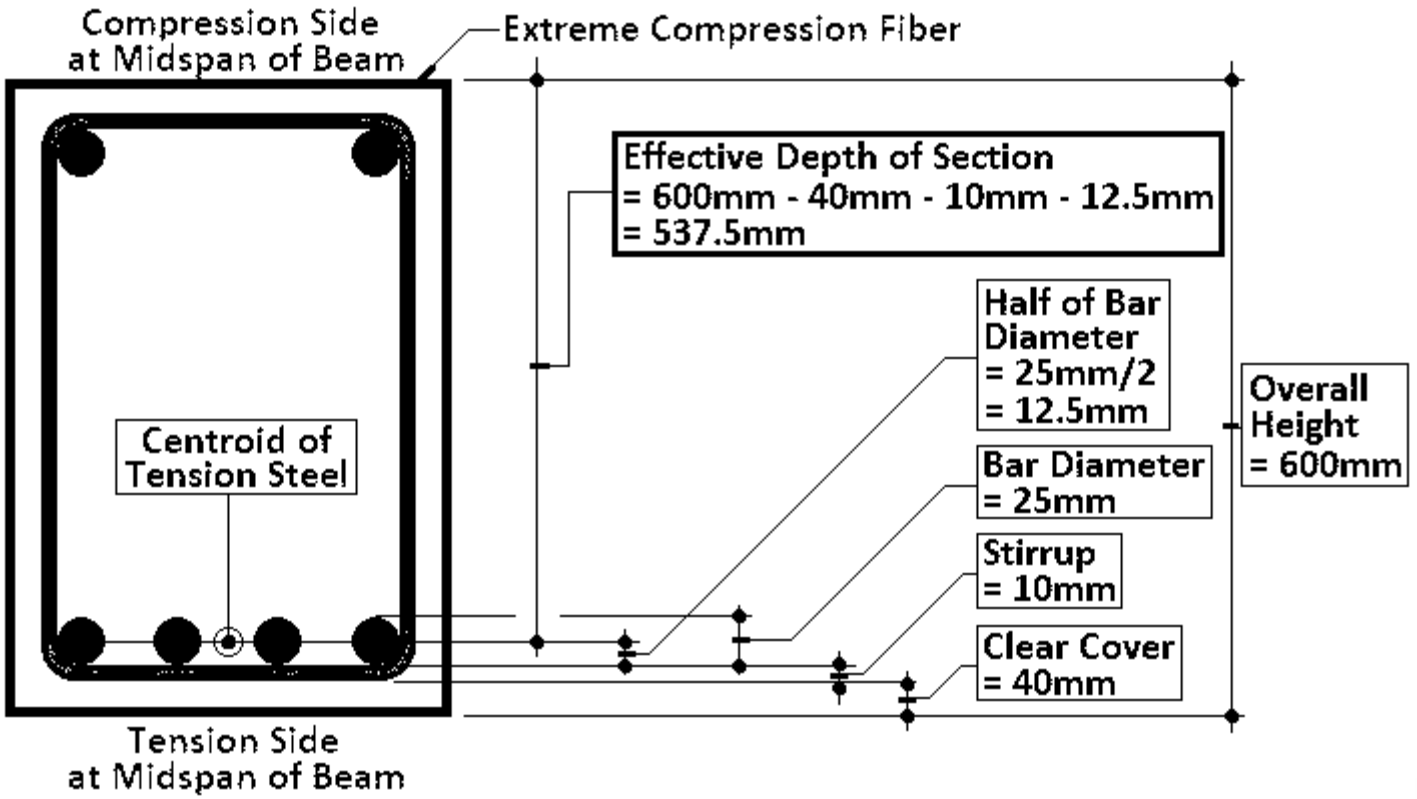
Height of Building	Quantity of Reinforcement in Kg/sq. m. of Plinth Area			Remarks
	Foundation	Floor Beams per floor	Floor Slabs per floor	
9 m to 13.5 m maximum	22 to 33	18 to 20	10 to 12	
13.5 m to 21 m maximum	50 to 65	do	do	
21 m to 36 m maximum	95 to 110	do	do	Foundation has to be raft slab. Piling will depend on soil condition.
Above 36 m	130 to 150	do	do	Concrete piling is generally required

Thumb Rule

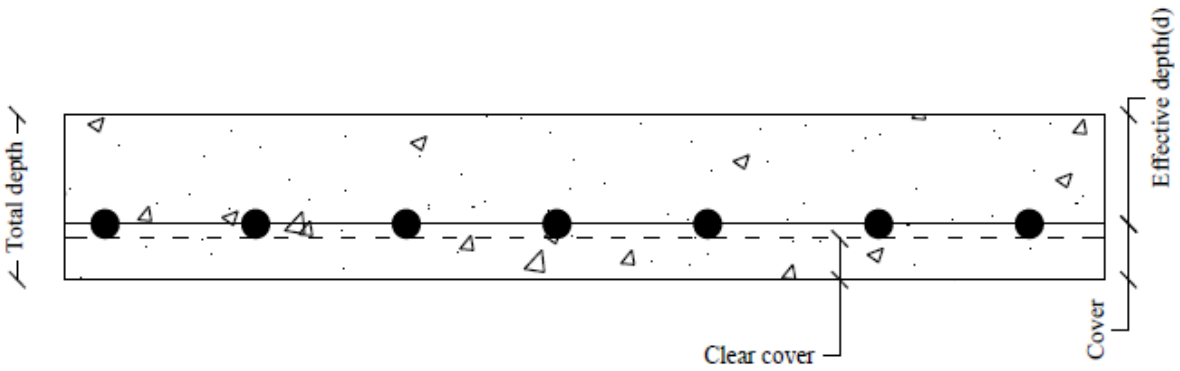
- A three storied building (each story being 10 feet) having a plinth area of 60 feet x 45 feet & height of 30 feet. Estimate the quantity of reinforcement in each floor & total reinforcement in the floors using thumb rule.

➢ Plinth Area, Quantity of Reinforcement in kg/sq. m. of plinth area, Number of floors

Accurate Estimate Based on Detailed Drawings



Accurate Estimate Based on Detailed Drawings



Total Depth (t) = Face to face distance
Effective Depth (d) = Distance from compression face of concrete to the center of gravity (C.G.) of steels
Cover = Distance of outer fiber from C.G. of steel
Clear Cover = Distance of outer fiber from face of steel
Total Depth (t) = Effective Depth (d) + Cover

Reinforcement Covering

- To protect reinforcement/steel from weathering effect i.e. corrosion
- To protect from fire.
- The need for adequate adhesion between the steel and concrete.

For normal exposure of concrete according to ACI Code (American Concrete Institute)

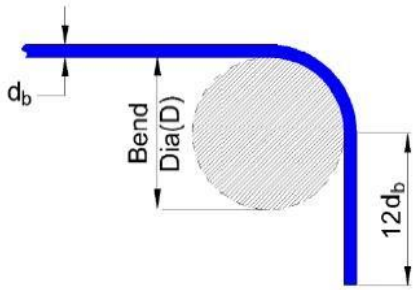
	Beam/Column	Slab
Clear cover	1.5 inch (minimum)	0.75 inch (minimum)
Cover	2.5 inch	1 inch

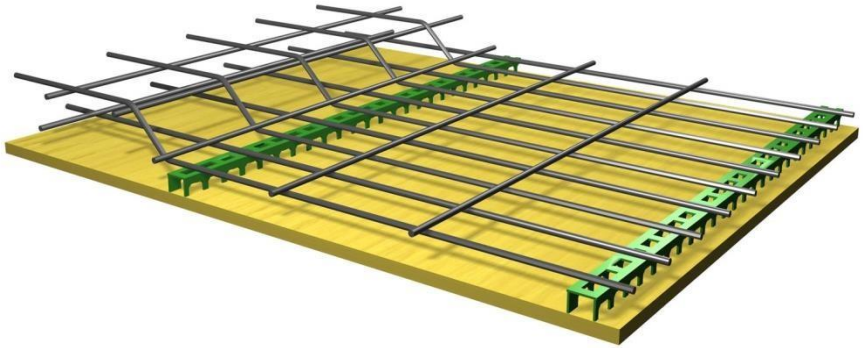
Reinforcement in Slab

- Straight bar : lies at bottom
- Cranked bar: lies at top & at bottom (both location, depending upon cracking)
- Extra top: lies at top

These three types of reinforcement occurs in both direction (both long and short direction).

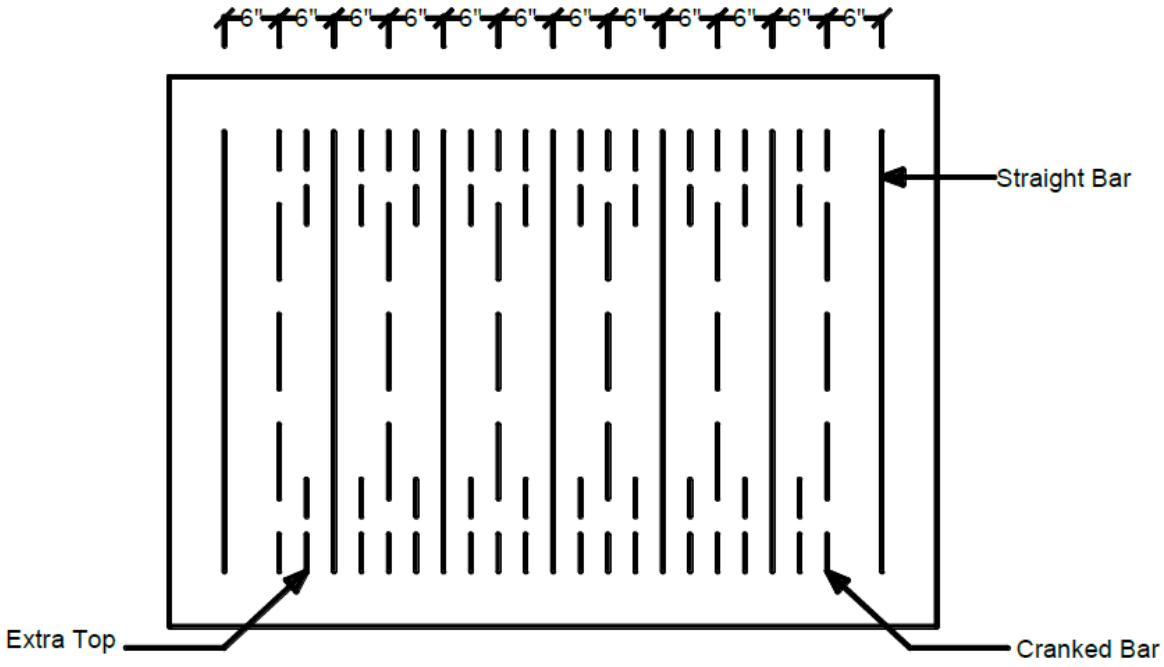
Each bar is hooked at each end if plain bar is used.





Slab Reinforcement Distribution

- Start with a straight bar and end with a straight bar
- Cranked bars are in between straight bars.
- Extra tops are in between cranked bar.



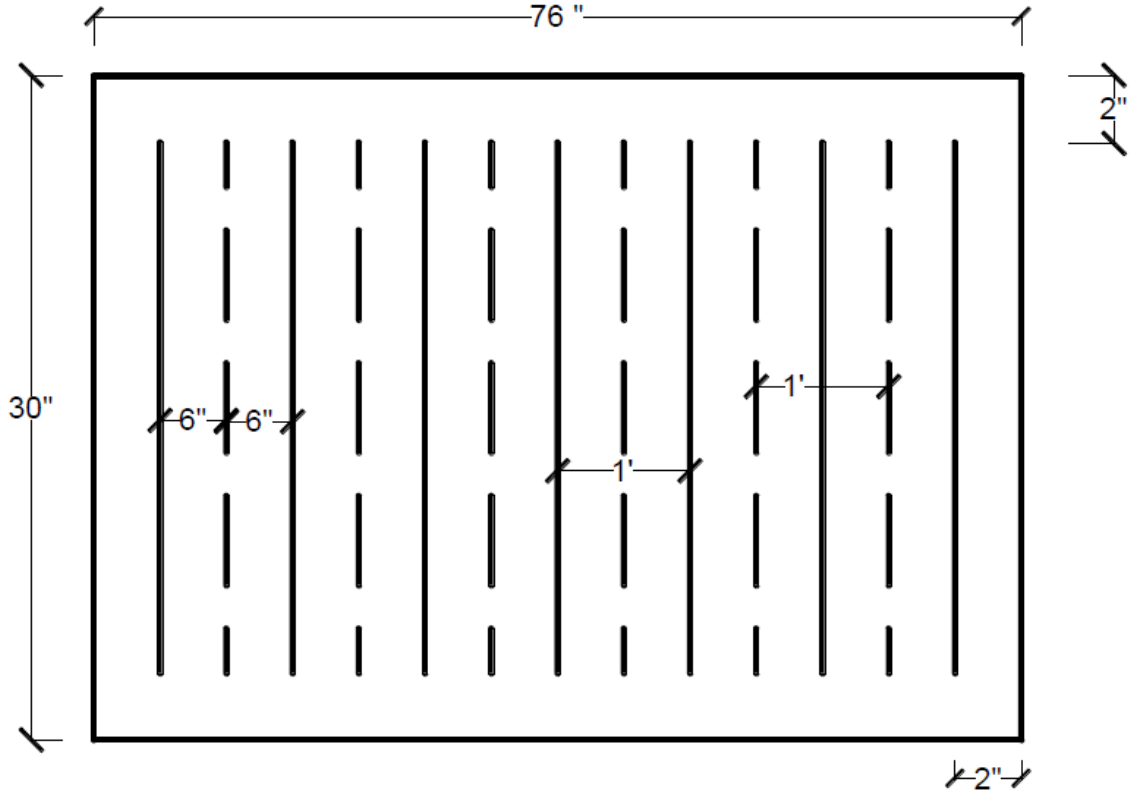
Slab Reinforcement Distribution

- Number of straight bar = $\frac{Distance\ -2"\ (from\ eac\square\ side)}{spacing\ of\ straig\square t\ bars} + 1$

= Next higher integer

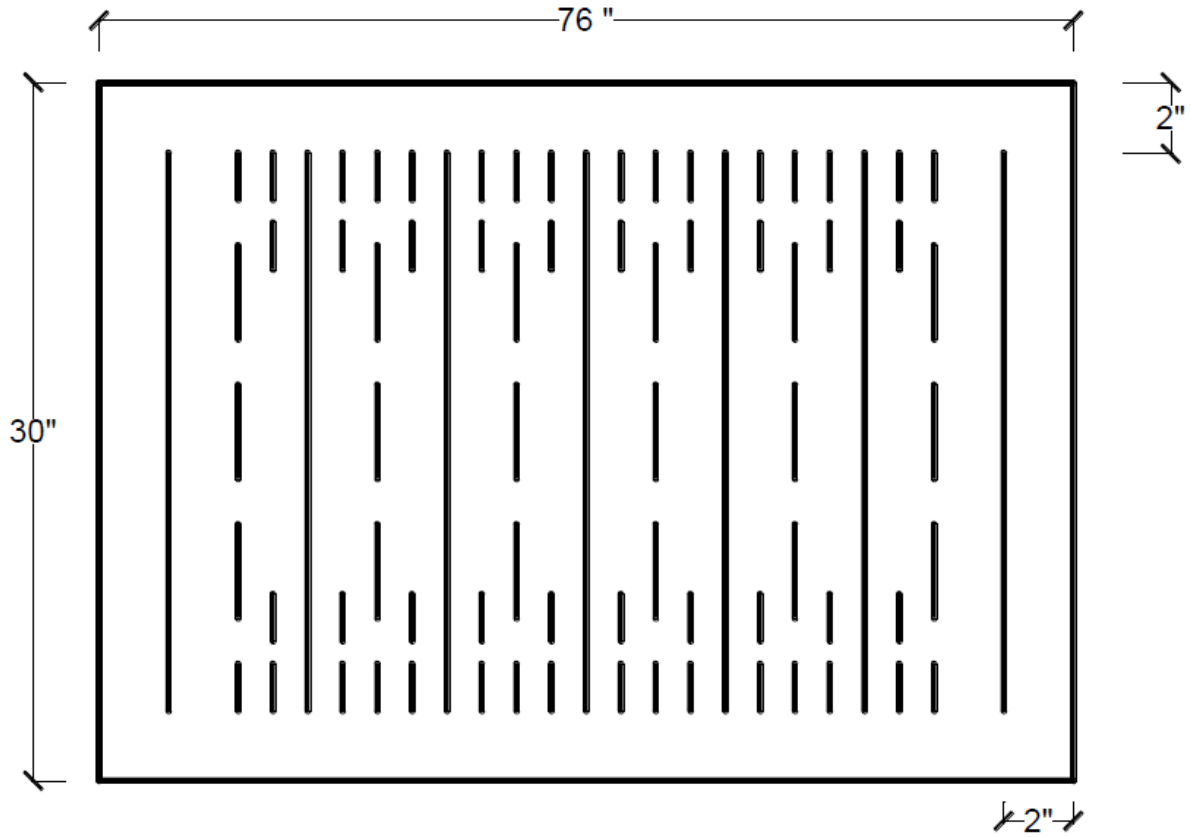
(distance is perpendicular to bar length)

- Between N number of straight bars, there will be (N-1) number of gaps. So, *No. of cranked bar = No. of straight bar -1*
- Between (N-1) number of cranked bars, there will be (N-2) number of gaps. So, *No. of extra top = No. of cranked bar – 1* (if one extra top in between cranked bars)
- No. of extra top = *n x (No. of cranked bar -1)* (if n extra tops in between cranked bars)



#3@6” c/c alt. ckd.

No. of straight bars = (76”- 2” - 2”)/12 + 1 = 6.16 = 7 Nos.
No. of cranked bars = No. of straight bars - 1 = 7 – 1 = 6 Nos.



2#3 extra top in between cranked

No of extra top (one side) = (No. of cranked bar – 1) x 2

= (6 – 1) x 2

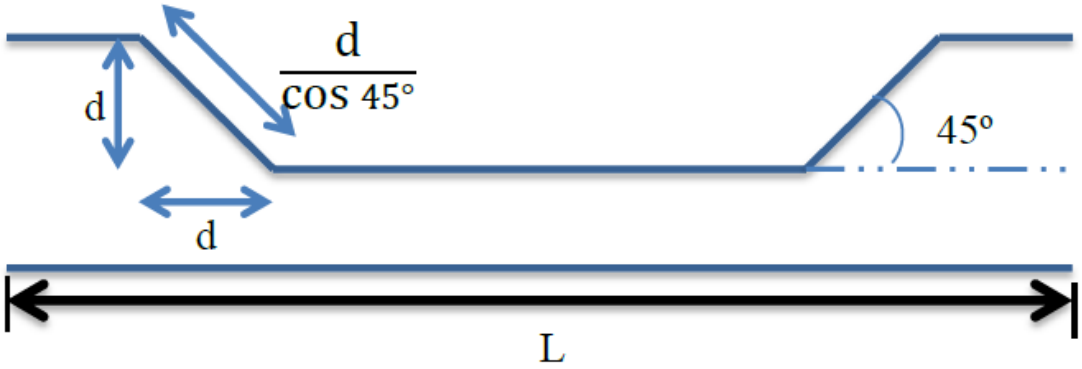
= 10

Length of bar

- Length of straight bar = Length of slab (L) – 2 inch (from each side)
- Length of cranked bar $= (L - d - d) + \frac{d}{\cos 45} + \frac{d}{\cos 45}$

$= (L - d - d) + 1.42d + 1.42d$

$= L + 0.42d + 0.42d$



Length of cranked bar = Length of straight bar + n (0.42d)

Where n is number of bends, d is effective depth of slab

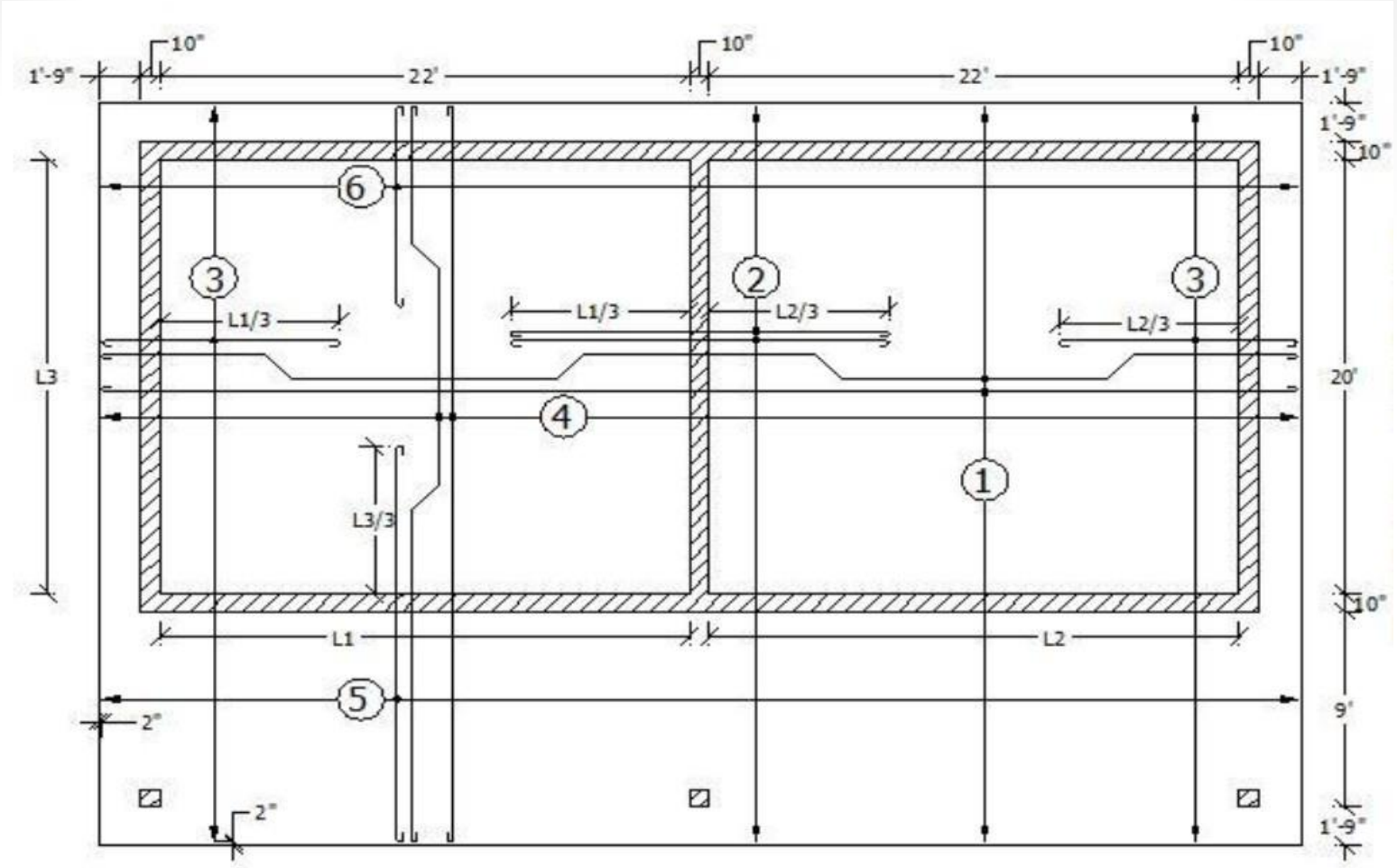
Length of hook

- The standard hook length for each end of plain rods vary from 9 to 12.5 times diameter of the rods generally used

- If plain bar is used, each bar must be hooked at both ends.
- If deformed bar is used, no hooks required.

Procedure for reinforcement estimation in slab

- Step 1: Determine the number of straight bars
- Step 2: Determine the number of cranked bar
- Step 3: Determine the number of extra top
- Step 4: Determine the number of hook
- Step 5: Determine the length of one straight bar
- Step 6: Determine the length of one cranked bar
- Step 7: Determine the length of extra top(s)
- Step 8: Determine the length of hook(s)
- Step 9: Determine the total length of straight bars
- Step 10: Determine the total length of cranked bars
- Step 11: Determine the total length of extra top(s)
- Step 12: Determine the total length of hook(s)
- Step 13: Repeat step 1 to step 12 for other direction
- Step 14: Volume of bar = Area of bar x Length of bar
- Step 15: Weight of bar = Unit weight of steel (490 pcf) x Volume



Legends:

(1) #4 @ 6" c/c alt. ckd

(2) 2#4 extra top in between ckd. bars

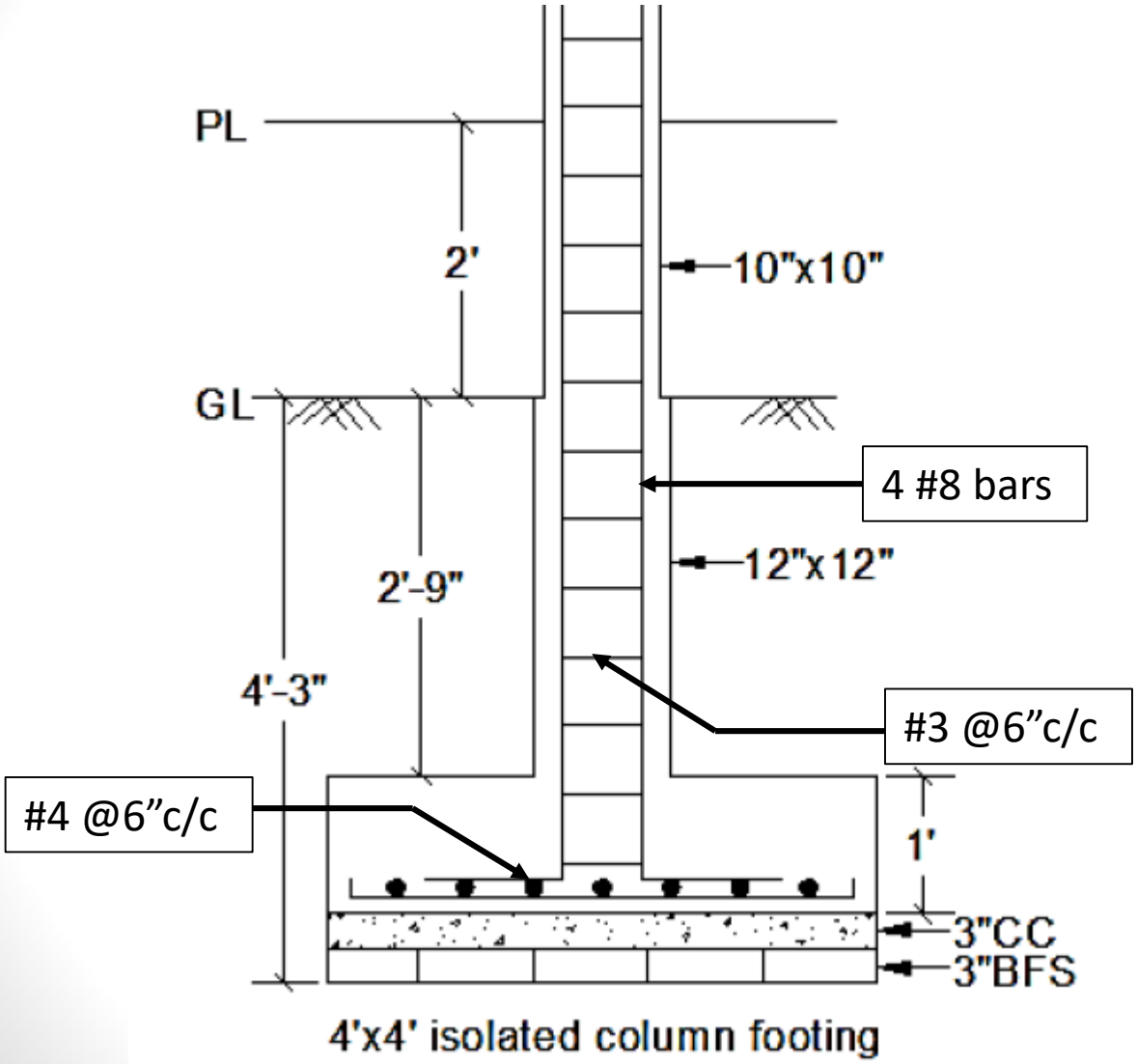
(3) 1#3 extra top in between ckd. Bars

(4) #3 @ 4" c/c alt. ckd

(5)1#4 extra top in between ckd. bars

(6)1#3 extra top in between ckd. bars

Reinforcement Estimation in Footing



Procedure for reinforcement estimation in footing

- Step 1: Determine the number of straight bars in base
- Step 2: Determine the length of one straight bar in base
- Step 3: Determine the total length of straight bars
- Step 4: Determine the number of hook(s)
- Step 5: Determine the length of hook(s)
- Step 6: Determine the total length of hook(s)
- Step 7: Repeat step 1 to step 6 for other direction
- Step 8: Determine the number of main bar
- Step 9: Determine the length of one main bar
- Step 10: Determine the total length of main bar
- Step 11: Determine the number of stirrups
- Step 12: Determine the length of one stirrup
- Step 13: Determine the total length of stirrups
- Step 14: Volume of bar = Area of bar x Length of bar
- Step 15: Weight of bar = Unit weight of steel (490 pcf) x Volume